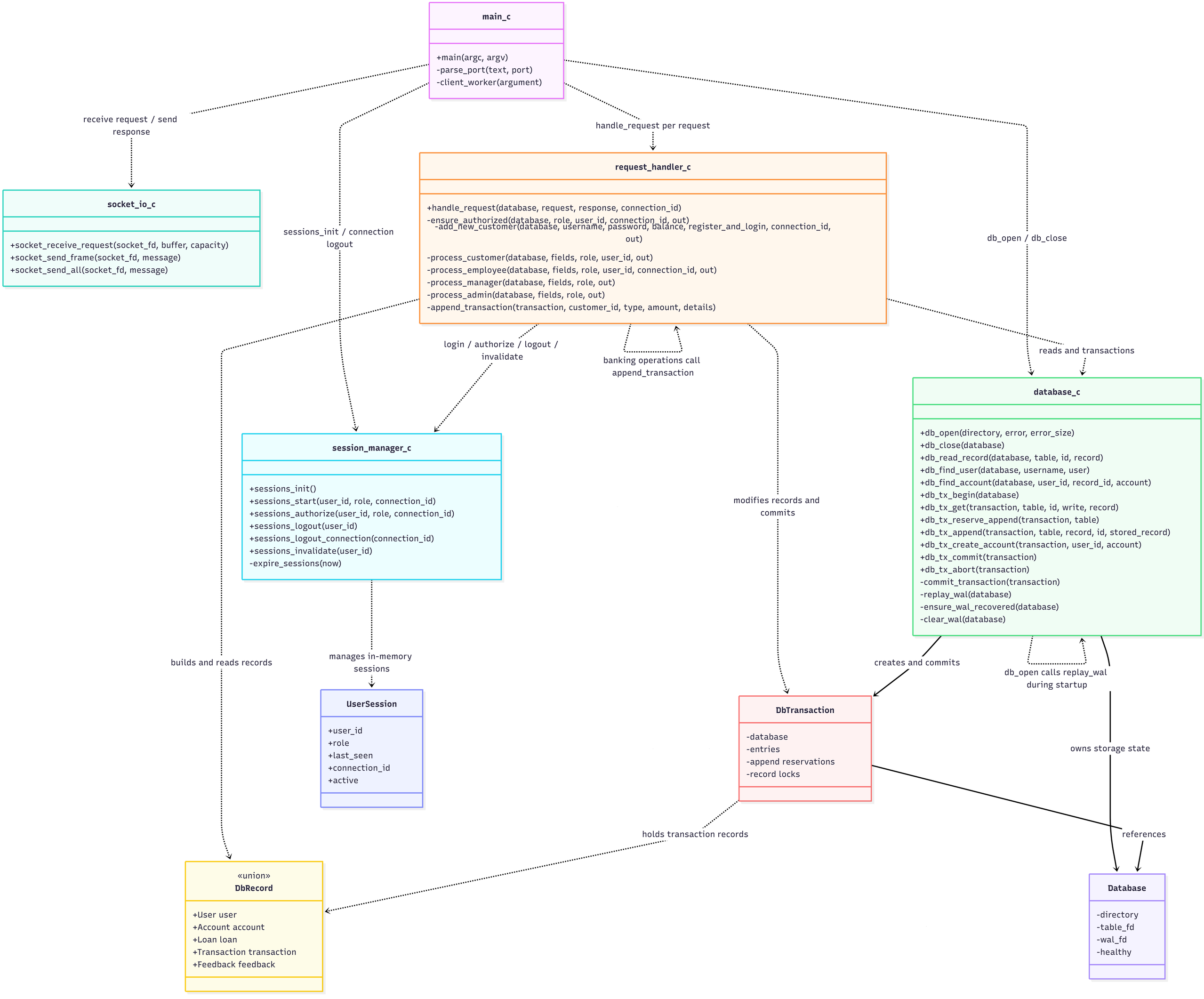

main_c
+main(argc, argv)
-parse_port(text, port)
-client_worker(argument)
receive request / send response
handle_request per request
sessions_init / connection logout
db_open / db_close
socket_io_c
+socket_receive_request(socket_fd, buffer, capacity)
+socket_send_frame(socket_fd, message)
+socket_send_all(socket_fd, message)
request_handler_c
+handle_request(database, request, response, connection_id)
-ensure_authorized(database, role, user_id, connection_id, out)
-add_new_customer(database, username, password, balance, register_and_login, connection_id, out)
-process_customer(database, fields, role, user_id, out)
-process_employee(database, fields, role, user_id, connection_id, out)
-process_manager(database, fields, role, out)
-process_admin(database, fields, role, out)
-append_transaction(transaction, customer_id, type, amount, details)
login / authorize / logout / invalidate
banking operations call append_transaction
reads and transactions
database_c
+db_open(directory, error, error_size)
+db_close(database)
+db_read_record(database, table, id, record)
+db_find_user(database, username, user)
+db_find_account(database, user_id, record_id, account)
+db_tx_begin(database)
+db_tx_get(transaction, table, id, write, record)
+db_tx_reserve_append(transaction, table)
+db_tx_append(transaction, table, record, id, stored_record)
+db_tx_create_account(transaction, user_id, account)
+db_tx_commit(transaction)
+db_tx_abort(transaction)
-commit_transaction(transaction)
-replay_wal(database)
-ensure_wal_recovered(database)
-clear_wal(database)
session_manager_c
+sessions_init()
+sessions_start(user_id, role, connection_id)
+sessions_authorize(user_id, role, connection_id)
+sessions_logout(user_id)
+sessions_logout_connection(connection_id)
+sessions_invalidate(user_id)
-expire_sessions(now)
modifies records and commits
builds and reads records
manages in-memory sessions
creates and commits
db_open calls replay_wal during startup
owns storage state
UserSession
+user_id
+role
+last_seen
+connection_id
+active
DbTransaction
-database
-entries
-append reservations
-record locks
holds transaction records
references
«union»
DbRecord
+User user
+Account account
+Loan loan
+Transaction transaction
+Feedback feedback
Database
-directory
-table_fd
-wal_fd
-healthy